

C797

Data Science and Analytics

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A hospital unit renovated in 2022 recorded lengths of stay (in days) for nine patients: 2, 2, 2, 3, 4, 4, 5, 6, 30. The unit has 28 beds. Which value best represents a typical stay?

- A. The mode, 2 days.
- B. The median, 4 days.
- C. The range, 28 days.
- D. The mean, about 6.4 days.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

A report shows that mean ED wait time is much higher than median wait time. What does that suggest?

- A. The distribution is right-skewed by a few very long waits.
- B. The mean was calculated wrong, since mean and median are always equal.
- C. The distribution is symmetric.
- D. The distribution is left-skewed.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

Systolic blood pressures in a wellness screening at a regional employer with a new gym are roughly normal, with a mean of 120 mmHg and a standard deviation of 10 mmHg. About what percentage of readings fall between 100 and 140?

- A. About 47.5%
- B. About 99.7%
- C. About 68%
- D. About 95%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

For the same readings (mean 120, SD 10), about what percentage are above 130?

- A. About 84%
- B. About 32%
- C. About 34%
- D. About 16%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A nurse researcher tests whether a new discharge checklist lowers readmissions. With alpha set at 0.05, the p-value is 0.03. The study ran at two hospitals over 18 months. What is the correct conclusion?

- A. Fail to reject the null hypothesis.
- B. There is a 3% chance the null hypothesis is true.
- C. Reject the null hypothesis; the result is statistically significant.
- D. The checklist is proven to work in every hospital.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A study at an urgent-care chain concludes that a new triage tool shortens wait times, but in reality the tool has no effect. The chain has 14 locations. Which error occurred?

- A. No error, since any significant result is correct.
- B. Type II error (false negative).
- C. Type I error (false positive).
- D. Sampling bias.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

A team wants to lower its chance of missing a real effect (Type II error) without changing alpha. What most directly helps?

- A. Lower alpha to 0.01.
- B. Increase the sample size.
- C. Report the results with only descriptive statistics.
- D. Nothing; Type II error cannot be changed.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A clinical analyst at a health system with a new data warehouse compares mean HbA1c across three diet programs with 40 patients in each. Which test fits?

- A. Three separate t-tests, each at alpha 0.05.
- B. Chi-square test.
- C. An independent t-test.
- D. One-way ANOVA.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

An analyst wants to know whether smoking status (yes/no) is associated with 30-day readmission (yes/no). Which test fits?

- A. Chi-square test of independence.
- B. One-way ANOVA.
- C. Paired t-test.
- D. Pearson correlation.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

The 95% confidence interval for mean weekly overtime among nurses at a hospital with a new parking garage is 4.2 to 6.8 hours. Which interpretation is correct?

- A. 95% of nurses work between 4.2 and 6.8 overtime hours.
- B. We are 95% confident the true mean overtime for all nurses there is between 4.2 and 6.8 hours.
- C. The true mean is definitely between 4.2 and 6.8 hours.
- D. There is a 95% chance the sample mean is in the interval.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A satisfaction survey at a pediatric clinic with a fish tank asks families to rate care as poor, fair, good, or excellent. What level of measurement is this?

- A. Interval.
- B. Nominal.
- C. Ordinal.
- D. Ratio.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A control chart of daily door-to-provider time at an ED near a major interstate shows points varying randomly within the control limits for 60 days, with no patterns. The ED director wants to investigate every day that lands above the mean. What is the best advice?

- A.** Treat it as common-cause variation; if the average is too high, change the process (for example, with a PDSA cycle).
- B.** Widen the control limits to 4 standard deviations.
- C.** Investigate each high day as a special cause.
- D.** Conclude the process is performing well because it is in control.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	____	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

A hospital unit ~~renovated in 2022~~ recorded lengths of stay (in days) for nine patients: 2, 2, 2, 3, 4, 4, 5, 6, 30. ~~The unit has 28 beds.~~ Which value best represents a typical stay?

Target word: typical value **Decides:** One 30-day outlier skews the data.

A	Adjacent category	The most frequent value sits at the low end, not the center.
B	Correct	The middle (5th) value resists the outlier.
C	Wrong stem number	Range measures spread, and it happens to match the bed count.
D	Confusion set	$58 / 9 \approx 6.4$, pulled up by the 30-day stay.

2. Answer A Concept 1 · transfer

A report shows that mean ED wait time is much higher than median wait time. What does that suggest?

Target word: reverse: infer shape from center **Decides:** Mean above median.

A	Correct	High outliers pull the mean above the median.
B	Absolute language	They differ whenever data are skewed.
C	Dropped element	In symmetric data the mean and median are about equal.
D	Inverted operation	Left skew puts the mean below the median.

3. Answer D Concept 2

Systolic blood pressures in a wellness screening ~~at a regional employer with a new gym~~ are roughly normal, with a mean of 120 mmHg and a standard deviation of 10 mmHg. About what percentage of readings fall between 100 and 140?

Target word: percentage in range **Decides:** 100 to 140 is ± 2 SD.

A	Dropped element	Covers only one side of the mean.
B	Adjacent category	That is ± 3 SD (90 to 150).
C	Confusion set	That is ± 1 SD (110 to 130).
D	Correct	± 2 SD covers about 95% of a normal distribution.

4. Answer D Concept 2 · transfer

For the same readings (mean 120, SD 10), about what percentage are above 130?

Target word: solve for one tail **Decides:** 130 is +1 SD; 68% lies within ± 1 SD.

A	Inverted operation	That is the percentage below 130.
B	Dropped element	Both tails combined; forgets to split in half.
C	Confusion set	That is between the mean and +1 SD.
D	Correct	$(100\% - 68\%) / 2 = 16\%$ in the upper tail.

5. Answer C Concept 3

A nurse researcher tests whether a new discharge checklist lowers readmissions. With alpha set at 0.05, the p-value is 0.03. ~~The study ran at two hospitals over 18 months.~~ What is the correct conclusion?

Target word: decision about the null **Decides:** $0.03 \leq 0.05$.

A	Inverted operation	That is the decision when p is greater than alpha.
B	Confusion set	p assumes the null is true; it is not the probability that it is true.
C	Correct	p is at or below alpha.
D	Absolute language	Significance does not prove the effect everywhere.

6. Answer C Concept 4

A study at an urgent-care chain concludes that a new triage tool shortens wait times, but in reality the tool has no effect. ~~The chain has 14 locations.~~ Which error occurred?

Target word: error type **Decides:** A true null was rejected.

A	Absolute language	Significant results can still be false positives.
B	Inverted operation	Type II is missing a real effect.
C	Correct	Claiming an effect that does not exist.
D	Adjacent category	Bias is a design flaw, not the name of this decision error.

7. Answer B Concept 4 · transfer

A team wants to lower its chance of missing a real effect (Type II error) without changing alpha. What most directly helps?

Target word: reverse: reduce Type II **Decides:** Power rises with sample size.

A	Inverted operation	Lowering alpha raises Type II error, and it changes alpha.
B	Correct	Larger samples raise power and cut Type II error.
C	Adjacent category	Avoids testing rather than improving it.
D	Absolute language	Sample size and effect size both affect it.

8. Answer D Concept 5

A clinical analyst ~~at a health system with a new data warehouse~~ compares mean HbA1c across three diet programs with 40 patients in each. Which test fits?

Target word: test choice **Decides:** Three group means, numeric outcome.

A	Dropped element	Drops control of the overall false-positive rate, which rises with each test.
B	Confusion set	For two categorical variables; HbA1c is numeric.
C	Adjacent category	Compares only two groups.
D	Correct	Compares means across three or more groups.

9. Answer A Concept 5 • transfer

An analyst wants to know whether smoking status (yes/no) is associated with 30-day readmission (yes/no). Which test fits?

Target word: test for two categorical variables **Decides:** Both variables are categories.

A	Correct	Tests association between two categorical variables.
B	Confusion set	Needs a numeric outcome across groups.
C	Adjacent category	Compares two related numeric measurements.
D	Dropped element	Drops the need for continuous variables.

10. Answer B Concept 6

The 95% confidence interval for mean weekly overtime among nurses ~~at a hospital with a new parking garage~~ is 4.2 to 6.8 hours. Which interpretation is correct?

Target word: CI interpretation **Decides:** The interval is about the population mean.

A	Confusion set	The interval is about the mean, not individuals.
B	Correct	A CI estimates the population parameter.
C	Absolute language	About 5% of such intervals miss the true value.
D	Adjacent category	The sample mean (5.5) is always at the center; the uncertainty is about the population mean.

11. Answer C Concept 7

A satisfaction survey at a pediatric clinic ~~with a fish tank~~ asks families to rate care as poor, fair, good, or excellent. What level of measurement is this?

Target word: measurement level **Decides:** Ordered labels with no equal spacing.

A	Adjacent category	Assumes equal spacing between ratings.
B	Dropped element	Drops the order among the ratings.
C	Correct	Categories have an order, but distances between them are not equal.
D	Confusion set	Needs equal spacing and a true zero.

12. Answer A Concept 8

A control chart of daily door-to-provider time at an ED ~~near a major interstate~~ shows points varying randomly within the control limits for 60 days, with no patterns. The ED director wants to investigate every day that lands above the mean. What is the best advice?

Target word: response to a stable chart **Decides:** Random variation within limits is common cause.

A	Correct	Reacting to single points is tampering; improvement comes from process change.
B	Adjacent category	Changing limits hides signals and does not improve care.
C	Confusion set	Nothing signals a special cause.
D	Absolute language	Stable does not mean good; it can be stably too slow.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	6	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	9	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	5	
Absolute language	Always, never, guarantee — where judgment applies.	6	
Confusion set	Every option is real; the test is exact role-matching.	9	
Camouflage	Answers a stem fact that was never the question.	0	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Center of skewed data	The median resists outliers; in right-skewed data the mean is pulled above the median.	Reporting the mean as typical when there are extreme values.	One or two very large values, such as long stays or waits.
2	Normal distribution and the empirical rule	About 68% of values fall within 1 SD of the mean, 95% within 2 SD, and 99.7% within 3 SD.	Using 68% for a 2-SD range, or forgetting that tails split in half.	A mean, an SD, and a range or cutoff.
3	p-values and significance	If $p \leq \alpha$, reject the null; p is the probability of results this extreme if the null were true, not the probability the null is true.	Reading $p = 0.03$ as a 3% chance the null is true.	A p-value next to an alpha.
4	Type I and Type II errors	Type I = rejecting a true null (false positive); Type II = missing a real effect (false negative); larger samples raise power.	Swapping the two, or lowering alpha to cut Type II error.	"Concluded it works, but it doesn't" or "missed a real effect."
5	Choosing a statistical test	Two group means: t-test; three or more means: ANOVA; two categorical variables: chi-square; two continuous variables: correlation or regression.	Running many t-tests for three groups.	Count the groups and check whether the outcome is numeric or categorical.
6	Confidence intervals	A 95% CI gives a range of plausible values for the population parameter; the method captures the true value about 95% of the time.	Saying 95% of individuals fall in the interval.	"95% confidence interval for the mean."
7	Levels of measurement	Nominal = categories; ordinal = ordered categories; interval = equal spacing without a true zero; ratio = equal spacing with a true zero.	Treating ratings like poor-excellent as interval.	Ranked labels without equal distances.
8	Common- vs. special-cause variation	Points randomly within control limits reflect common-cause variation; reacting to each point is tampering; improve the process instead.	Investigating every above-average day, or calling a stable process good.	A control chart with no rule violations.

Course scope (public description): This course addresses the interdisciplinary and emerging field of data science in healthcare. Candidates learn to combine tools and techniques from statistics, computer science, data visualization, and the social sciences to solve problems using data. Topics include data analysis; database management; inferential and descriptive statistics; statistical inference; and process improvement.